PREDNISOLONE

Morning dose with food

Taper if >3 weeks

CHEMICAL NAME 11β,17,21-trihydroxypregna-1,4-diene-3,20-dione
CLASS Corticosteroid (intermediate-acting glucocorticoid)
TRADE NAMES Deltacortril · Pred Forte (eye) · Predsol · Millipred

THERAPEUTIC USES / INDICATIONS

- Asthma and COPD exacerbations.
- Inflammatory and autoimmune diseases: rheumatoid arthritis, SLE, vasculitis, polymyalgia rheumatica, giant cell arteritis, IBD flare, autoimmune hepatitis.
- Nephrotic syndrome, allergic reactions, skin disease.
- Haematological malignancies (with chemotherapy); croup (alternative); eye drops for anterior uveitis.

ADMINISTRATION

- **PO** once daily in the morning with food (mimics cortisol rhythm, ↓ adrenal suppression, ↓ dyspepsia).
- Soluble or enteric-coated tablets; oral solution for children.
- Eye drops: shake well.

DOSAGES

- **Acute asthma (adult):** 40–50 mg daily for 5 days (no taper needed).
- **COPD exacerbation:** 30–40 mg daily for 5 days.
- **Children (acute asthma):** <2 yr 10 mg, 2–5 yr 20 mg, >5 yr 30–40 mg daily, usually up to 3 days.
- **Polymyalgia:** 15 mg daily, slowly tapered over 1–2 years.
- **Nephrotic syndrome (child):** 60 mg/m²/day.

PHARMACOKINETICS

- **Onset:** 1–2 h. **Duration:** 12–36 h (intermediate).
- Active (prednisone is its prodrug); metabolised in liver (CYP3A4).
- Excreted in urine. **t½:** 2–4 h (biological 12–36 h).

PHARMACODYNAMICS

- **Glucocorticoid receptor** agonist → suppresses inflammatory gene transcription (cytokines, prostaglandins, leukotrienes), immune cell activity.
- Potency ~4× hydrocortisone; minor mineralocorticoid action.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- ↑ appetite, weight gain, dyspepsia.
- Insomnia, mood swings.
- Hyperglycaemia, fluid retention.

ADVERSE EFFECTS (SERIOUS)

- Adrenal suppression; adrenal crisis on abrupt withdrawal.
- Osteoporosis, avascular necrosis; myopathy.
- Diabetes, hypertension, cataract, glaucoma, infection (incl. TB reactivation, chickenpox), peptic ulcer.
- Cushingoid features; psychosis; growth suppression in children.

CONTRAINDICATIONS & CAUTIONS

- Systemic infection without cover; live vaccines at immunosuppressive doses.
- **Caution:** diabetes, osteoporosis, peptic ulcer, TB history, glaucoma, psychiatric illness.

KEY INTERACTIONS

- **Rifampicin, phenytoin, carbamazepine:** ↓ effect.
- **NSAIDs, aspirin:** GI bleeding.
- **Diuretics, β2 agonists:** hypokalaemia.
- **Antidiabetics, antihypertensives:** ↓ effect; **live vaccines:** avoid.

NURSING MANAGEMENT

- Give in the morning with food.
- Monitor glucose, BP, weight, mood, K^+.
- Long-term: bone protection (calcium/vitamin D, bisphosphonate), eye checks, gastroprotection when indicated.
- Plan gradual taper if used >3 weeks or high doses.
- Assess infection signs (may be masked).

PATIENT EDUCATION

- Take in the morning with breakfast.
- Do not stop suddenly after long courses — follow the taper.
- Carry a steroid card; avoid contact with chickenpox/shingles if not immune.
- Report black stools, mood changes, high glucose, signs of infection.

KEY POINTS

- Oral glucocorticoid; 4× hydrocortisone.
- Morning with food.
- Asthma: 40–50 mg × 5 days.
- Taper after >3 weeks — adrenal suppression.

SOURCES CHECKED **[639]** Prednisolone 1 mg Gastro-resistant Tablets (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158)

Full references p. 432–457